

Sharp total variation bounds from spherical rearrangement

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Abstract

We study the sharp control of total variation by a higher derivative under homogeneous endpoint conditions. For every integer $n \geq 2$, we prove that the norm of $u \mapsto u'$ from $W_0^{n,2}(0,1)$ to $L^1(0,1)$ is twice the norm of point evaluation into $L^\infty(0,1)$. The extremizers of the two embeddings coincide and are symmetric about the midpoint. This proves the Hilbert-space case of a conjecture of Nazarov and Shcheglova. The main step is a rearrangement principle for a clamped Green operator after projection onto the zero-mean subspace. A Jacobi-polynomial representation identifies its quadratic form with a positive mixture of Poisson interactions on a sphere, where rearrangement reduces the optimization to a cap. We also prove the constant identity and coincidence of maximizing sets at the measure endpoint $p = 1$, using the geometry of extreme moment measures and unimodal splines. For each fixed derivative order, we further prove the full variation identity and midpoint-symmetric equality classification on an open interval of exponents around two. A dual optimality equation gives compactness through the highest derivative; polynomial sublevel bounds control the point-evaluation profile on both sides of the Hilbert exponent. At the measure endpoint in derivative order four, we also determine the exact sharp constant and unique symmetric extremizer by a cubic alternation argument and an explicit algebraic exclusion of asymmetric maximizers. An all-order contact-motion argument then proves strict midpoint maximization and the unique symmetric measure extremizer in every order, completing the measure-endpoint conjecture. A weighted zero-count extension proves strict midpoint point-evaluation maximization and unique symmetric height extremizers for every finite exponent. This completes the function-height case of the broader symmetry conjecture when combined with the endpoint results. Finally, compactness of arbitrary variation maximizers at the unique measure endpoint proves the full variation statement on an order-dependent interval immediately above one. The full all-exponent variation conjecture remains open.

1. The sharp inequality

All functions and measures in this paper are real. For $1 \leq p \leq \infty$, we use the homogeneous norm $\|u^{(n)}\|_p$ on $W_0^{n,p}(0,1)$. The endpoint conditions are

$$u^{(j)}(0) = u^{(j)}(1) = 0, \quad 0 \leq j < n. \quad (1)$$

Let $C_{n,p}$ and $V_{n,p}$ denote the sharp constants in

$$\|u\|_\infty \leq C_{n,p} \|u^{(n)}\|_p, \quad \|u'\|_1 \leq V_{n,p} \|u^{(n)}\|_p. \quad (2)$$

The inequality $V_{n,p} \geq 2C_{n,p}$ follows from the variation of a function that vanishes at both endpoints. Nazarov and Shcheglova [NS, Conjecture 4.14] conjecture equality for all $n \geq 2$ and $1 \leq p \leq \infty$, together with coincidence and midpoint symmetry of the extremizers. Their Remark 4.15 records the previously known case $n = 2$.

Theorem 1. For every integer $n \geq 2$,

$$V_{n,2} = 2C_{n,2} = \frac{1}{2^{2n-2}(n-1)!\sqrt{2n-1}}. \quad (3)$$

The nonzero extremizers are precisely the scalar multiples of the Riesz representer $R_n(\cdot, 1/2)$ of evaluation at $1/2$ in $W_0^{n,2}(0,1)$. In particular, the maximizing sets in (2) coincide, and their elements are even about $1/2$.

The point-evaluation formula used here is prior work. Kalyabin [K, Theorem 2] gives, after affine rescaling,

$$\|\text{ev}_a\|^2 = \frac{[a(1-a)]^{2n-1}}{(2n-1)((n-1)!)^2}. \quad (4)$$

Our contribution is the upper bound for the variation norm and its equality classification, obtained through the rearrangement principle in Section 3. Best polynomial approximation already occurs in this embedding problem [GS]. The sharp $W_0^{m,2}$ to L^1 embedding of the function itself was proved by Hindov, Nitzan, Olsen and Rydhe [HNOR]; their theorem does not include the additional zero-mean constraint on a derivative that appears below.

2. A Jacobi representation of the clamped Green form

Fix $m \geq 1$, and set

$$w(x) = x^m(1-x)^m, \quad M = \int_0^1 w(x) dx = \frac{(m!)^2}{(2m+1)!}, \quad d\pi(x) = M^{-1}w(x) dx. \quad (5)$$

Let h_k be the orthonormal polynomials for π , with positive leading coefficients. These are normalized shifted Jacobi polynomials with parameters (m, m) ; in particular $h_0 = 1$. Put

$$v_k = wh_k, \quad \lambda_k = \frac{(k+2m)!}{k!}. \quad (6)$$

Lemma 2. The functions $v_k/\sqrt{M\lambda_k}$ form an orthonormal basis of $H = W_0^{m,2}(0,1)$ in its energy inner product. Moreover,

$$(-1)^m v_k^{(2m)} = \lambda_k h_k. \quad (7)$$

Proof. The left side of (7) has degree k . If q is a polynomial of degree less than k , integration by parts $2m$ times yields

$$\int_0^1 (wh_k)^{(2m)} wq = \int_0^1 wh_k (wq)^{(2m)} = 0. \quad (8)$$

Every boundary term vanishes because one of its two factors has fewer than m derivatives and retains an endpoint zero. Orthogonality therefore shows that the left side of (7) is proportional to h_k . Comparing leading coefficients proves (7). A further integration by parts gives

$$\langle v_k, v_\ell \rangle_H = M \lambda_k \delta_{k\ell}. \quad (9)$$

If $y \in H$ is orthogonal to every v_k , weak integration by parts gives $\int_0^1 y h_k = 0$ for every k . The h_k span the polynomials, which are dense in $C[0, 1]$, so $y = 0$. This proves completeness. $\square$

The mean functional $y \mapsto \int_0^1 y$ has Riesz representer $v_0/\lambda_0 = w/(2m)!$. Therefore

$$H_* := \left\{ y \in H : \int_0^1 y = 0 \right\} \quad (10)$$

is obtained by removing the zeroth basis vector. For a bounded real load g , Parseval gives the squared dual norm of $y \mapsto \int gy$ on H_* :

$$Q_m(g) = M \sum_{k \geq 1} \frac{\left| \int_0^1 g h_k d\pi \right|^2}{\lambda_k}. \quad (11)$$

Equivalently, if G_m is the clamped Green kernel and $w_m = w/(2m)!$, then

$$Q_m(g) = \iint g(x) G_m(x, t) g(t) dx dt - \frac{(\int g w_m)^2}{\int w_m}. \quad (12)$$

Formula (11) is sufficient for the proof and avoids any pointwise convergence issue for a Green series.

3. Spherical rearrangement of the load

Set $d = 2m + 2$ and let σ be normalized surface measure on S^d . The coordinate $x = (1 + \xi_{d+1})/2$ has law π . Thus

$$F(\xi) = g\left(\frac{1 + \xi_{d+1}}{2}\right) \quad (13)$$

identifies $L^2(\pi)$ with the zonal subspace of $L^2(S^d)$. Under this identification, h_k is the normalized zonal harmonic of degree k . This follows from the orthogonal decomposition of polynomials into spherical harmonics: a zonal polynomial of degree k orthogonal to all lower zonal degrees has only its degree- k component.

The spherical Poisson operator P_r , $0 < r < 1$, has eigenvalue r^k on degree- k harmonics and kernel

$$K_r(\xi \cdot \eta) = \frac{1 - r^2}{(1 - 2r\xi \cdot \eta + r^2)^{(d+1)/2}}. \quad (14)$$

We use normalized surface measure, as in [ABR, (1.15)]. The eigenvalue follows by harmonic extension of a homogeneous harmonic polynomial. The beta integral gives

$$\frac{1}{\lambda_k} = \frac{1}{(2m-1)!} \int_0^1 r^k (1-r)^{2m-1} dr. \quad (15)$$

Tonelli's theorem applied to the nonnegative spectral terms in (11) therefore gives

$$Q_m(g) = \frac{M}{(2m-1)!} \int_0^1 (1-r)^{2m-1} \left(\langle F, P_r F \rangle - \left(\int F d\sigma \right)^2 \right) dr. \quad (16)$$

The deletion of the constant harmonic is exactly the mean subtraction needed in (12).

Theorem 3. If $A \subset (0, 1)$ is measurable and $a \in [0, 1]$ satisfies $\pi(A) = \pi([0, a])$, then

$$Q_m(\mathbf{1}_A) \leq Q_m(\mathbf{1}_{[0,a]}). \quad (17)$$

If $0 < \pi(A) < 1$, equality holds only when A agrees, up to null sets, with an interval abutting one endpoint and having that π -measure.

Proof. Lift A to the zonal subset $E \subset S^d$ using (13). For fixed r , K_r is positive, bounded and strictly decreasing in geodesic distance. Spherical rearrangement [BH, Theorem 2] maximizes the self-interaction of $\mathbf{1}_E$ at fixed surface measure by a cap. Choose that cap about the negative coordinate axis. It corresponds to $[0, a]$. The mean term in (16) is unchanged, and integration proves (17).

For clarity, the equality hypotheses in [BH] are satisfied with two functions and integrand $(s, t) \mapsto st$. Its mixed finite difference is strictly positive; the kernel is positive and strictly decreasing; and the nontrivial indicators are nonconstant with finite interaction. Hence equality at fixed r forces E to be a cap up to rotation. If equality holds in (17), the nonnegative rearrangement deficit has zero integral in (16), so equality holds for almost every r , and in particular for one $r \in (0, 1)$. Thus E is a cap.

Its axis must be the original coordinate axis up to reversal. Indeed, zonal invariance makes the centroid of E parallel to that axis. A nontrivial cap has a nonzero centroid parallel to its own axis. Consequently A is an endpoint interval, as claimed. $\square$

The same argument proves a broader load rearrangement statement. For a bounded real g , let $g^\downarrow$ be its nonincreasing equimeasurable rearrangement relative to π . Adding a constant makes g nonnegative without changing Q_m . Applying [BH] in (16) gives

$$Q_m(g) \leq Q_m(g^\downarrow). \quad (18)$$

This is rearrangement of a load in a Green quadratic form. It does not assert that ordinary rearrangement preserves a higher derivative norm.

4. Proof of Theorem 1

Put $m = n - 1$. The map $u \mapsto y = u'$ is an isometric bijection from $W_0^{n,2}$ to H_* : integrating y recovers u , and the zero-mean condition supplies $u(1) = 0$.

Duality gives

$$V_{n,2}^2 = \sup_{|g| \leq 1} Q_m(g) = 4 \sup_A Q_m(\mathbf{1}_A). \quad (19)$$

To justify the second equality, write a bounded g with $|g| \leq 1$ as the average of $\text{sign}(g - s)$ over uniform $s \in [-1, 1]$. Convexity of the squared Hilbert dual norm bounds its value by the supremum over sign functions. Constants are annihilated in (11), giving $Q_m(2\mathbf{1}_A - 1) = 4Q_m(\mathbf{1}_A)$.

For an endpoint interval,

$$\int_0^a y(x) dx = u(a), \quad Q_m(\mathbf{1}_{[0,a]}) = \|\text{ev}_a\|^2. \quad (20)$$

Theorem 3, (4), and (19) now prove (3). In particular, the maximizing interval is a half-interval.

Suppose that $u \neq 0$ attains equality, and take $g = \text{sign}(u')$, choosing either sign where $u' = 0$. Both signs occur on sets of positive measure. Equality in the Hilbert dual bound and in Theorem 3 forces g to be a half-interval sign function, up to its global sign and null sets. Its functional on H_* is $\pm 2u(1/2)$. Equality in Cauchy--Schwarz therefore makes u a scalar multiple of $R_n(\cdot, 1/2)$.

Conversely, this representer maximizes point evaluation. Its variation is at least twice its maximum and at most the bound just proved, so it also maximizes variation. Formula (4) has its unique maximum at $a = 1/2$, and uniqueness of a Hilbert-space Riesz representer gives precisely the same classification for the point-norm extremizers. Finally, reflection preserves the energy inner product and fixes evaluation at $1/2$, so uniqueness makes $R_n(\cdot, 1/2)$ reflection invariant. $\square$

5. The measure endpoint

Let $\mathcal{M}_n$ consist of real finite signed measures μ on $[0, 1]$ such that

$$\|\mu\|_{\text{TV}} \leq 1, \quad \int_0^1 t^j d\mu(t) = 0 \quad (0 \leq j < n). \quad (21)$$

Their compactly supported potentials are

$$U_\mu(x) = \int_0^1 \frac{(x-t)_+^{n-1}}{(n-1)!} d\mu(t). \quad (22)$$

The moments make U_μ vanish off $[0, 1]$, and $D^n U_\mu = \mu$. For $n \geq 2$, U_μ is continuous and absolutely continuous, with integrable first derivative. Endpoint atoms are allowed in this relaxed formulation.

Theorem 4. For every $n \geq 2$,

$$\sup_{\mu \in \mathcal{M}_n} \|U'_\mu\|_1 = 2 \sup_{\mu \in \mathcal{M}_n} \|U_\mu\|_\infty. \quad (23)$$

The maximizing measures for the two quantities coincide. The relaxed constants equal $V_{n,1}$ and $C_{n,1}$, respectively. No symmetry assertion at $p = 1$ is included.

Proof of the constant identity. Every extreme point of $\mathcal{M}_n$ has norm one and exactly $n+1$ support points. To prove the support bound, suppose μ has norm one and at least $n+2$ disjoint Borel sets of positive $|\mu|$ -measure. A nonzero function h constant on those sets can satisfy the n equations $\int t^j h d\mu = 0$ and the extra equation $\int h d|\mu| = 0$. Then $(1 \pm \varepsilon h)\mu$ are distinct norm-one admissible measures for small ε . Such a μ is not extreme. A nonzero measure of norm less than one is excluded by scalar perturbation. Zero is the midpoint of opposite nonzero moment measures, which exist on any $n+1$ distinct nodes, and is not extreme either. A nonzero measure on at most n points is excluded by the Vandermonde determinant.

On nodes $t_0 < \dots < t_n$, the moment nullspace is one-dimensional, with weights

$$b_i = \prod_{j \neq i} (t_i - t_j)^{-1}. \quad (24)$$

Thus every extreme measure is a normalized signed multiple of $\sum_i b_i \delta_{t_i}$. Its potential is a B-spline, up to scale and sign. More explicitly, the Genocchi--Hermite identity [dB, (52)] identifies

$$\rho(x) = n \sum_{i=0}^n b_i (t_i - x)_+^{n-1} = (-1)^n n! U_{\sum b_i \delta_{t_i}}(x) \quad (25)$$

with the density of $\sum_i t_i X_i$, where X is uniform on the standard n -simplex. After parallel sections of the simplex are identified by translation, their Minkowski interpolation is contained in the section at the interpolated coordinate. The classical Brunn--Minkowski inequality [GHW, (3), $p = 1$] therefore makes their volumes to the power $1/(n-1)$ concave. The coarea factor relating section volume to ρ is constant, so $\rho^{1/(n-1)}$ is concave and ρ is unimodal. It is positive between its end knots, continuous, and zero at those knots. Every extreme potential therefore satisfies

$$\|U'_\mu\|_1 = 2\|U_\mu\|_\infty. \quad (26)$$

Write $C = \sup_{\mathcal{M}_n} \|U_\mu\|_\infty$. The set $\mathcal{M}_n$ is weak-star compact. Moreover, $\mu \mapsto \|U'_\mu\|_1$ is weak-star lower semicontinuous: it is the supremum of the continuous linear functionals $\int H_g d\mu$, where $|g| \leq 1$ and

$$H_g(t) = \int_t^1 \frac{(x-t)^{n-2}}{(n-2)!} g(x) dx. \quad (27)$$

Its sublevel set at $2C$ is closed and convex and contains all extreme points by (26). Krein--Milman therefore gives the upper bound in (23). The elementary variation lower bound gives the reverse inequality. The height supremum is attained: on the bounded measure set, joint continuity of the kernels in (22) makes the potential map weak-star to uniform continuous. Consequently the variation supremum is attained as well.

Coincidence. Let μ maximize variation, choose $g = \text{sign}(U'_\mu)$, and let $\mathcal{F}$ be the compact exposed face on which $\int H_g d\nu = 2C$. Its extreme points are extreme in $\mathcal{M}_n$. For each such point e ,

$$2C = \int g U'_e \leq \|U'_e\|_1 = 2\|U_e\|_\infty \leq 2C. \quad (28)$$

Thus g agrees with the sign of U'_e wherever the latter is nonzero, and every U_e maximizes height. Its end knots must be 0 and 1: a potential supported on a shorter interval of length L can be stretched to $[0, 1]$, keeping its derivative-measure norm one and multiplying its height by $L^{1-n} > 1$.

All U_e have the same global sign, since opposite signs give opposite nonzero derivatives on a common interval near zero, contradicting their agreement with g . Change that common sign to positive. Concavity of $U_e^{1/(n-1)}$ makes its maximum set a closed interval, with strictly positive derivative almost everywhere before it and strictly negative derivative almost everywhere after it. Two disjoint maximum intervals would again force contradictory signs for g . The maximum intervals therefore intersect pairwise, and hence have a common point a_* . All $U_e(a_*) = C$. Evaluation is weak-star continuous, so Krein--Milman applied to $\mathcal{F}$ gives $U_\mu(a_*) = C$. Conversely, a height maximizer is a variation maximizer by the already proved constant identity.

Recovery of the ordinary constants. For an ordinary admissible u , the measure $u^{(n)} dx$ satisfies (21). Conversely, mollify the zero extension of U_μ with a nonnegative mollifier supported in $[-\varepsilon, \varepsilon]$, obtaining U_ε . With $a = 1 + 2\varepsilon$, set

$$u_\varepsilon(x) = a^{1-n} U_\varepsilon(ax - \varepsilon). \quad (29)$$

It is smooth, supported on $[0, 1]$, satisfies all endpoint conditions, and has $\|u_\varepsilon^{(n)}\|_1 \leq 1$. Its maximum and variation converge to those of U_μ , by uniform convergence of the potential mollification and L^1 convergence of its first derivative. This proves equality of the relaxed and ordinary sharp constants. The proof classifies coincidence of relaxed maximizers and does not assert attainment in the ordinary $W_0^{n,1}$ class. $\square$

6. An open interval of exponents

The Hilbert theorem also yields a neighborhood of exponents for each fixed derivative order. The interval is not asserted to be uniform in the order. In this section write $X_{n,p} = W_0^{n,p}(0, 1)$ for the clamped space defined by (1), with norm $\|u^{(n)}\|_p$.

Theorem 5 (local continuation in the exponent). For every fixed integer $n \geq 3$, there exists $0 < \varepsilon_n < 1$ such that, if $|p - 2| < \varepsilon_n$, then

$$V_{n,p} = 2C_{n,p}. \quad (6.1)$$

All nonzero extremizers in the two inequalities are the scalar multiples of the unique positive norm-one extremizer for evaluation at $1/2$. They are symmetric about the midpoint and, after a choice of global sign, have strictly positive derivative on $(0, 1/2)$ and strictly negative derivative on $(1/2, 1)$.

The radius is allowed to depend on n and is not estimated here. The theorem does not reach all $1 < p < \infty$, give an interval uniform in n , or settle the full all-exponent variation conjecture.

6.1. Dual representation and compactness of variation maximizers

Let $\mathcal{P}_{n-1}$ denote the polynomials of degree at most $n - 1$. Integration from the left endpoint identifies $X_{n,p}$ isometrically with

$$Z_{n,p} = \{f \in L^p(0, 1) : \int_0^1 t^j f(t) dt = 0, \ 0 \leq j < n\}. \quad (6.2)$$

Indeed $u(x) = \int_0^x (x - t)^{n-1} f(t)/(n - 1)! dt$; the right endpoint conditions are equivalent to the displayed moments. For bounded g , put

$$H_g(t) = \int_t^1 \frac{(x - t)^{n-2}}{(n - 2)!} g(x) dx. \quad (6.3)$$

Fubini gives $\int g u' = \int H_g f$. If $q = p/(p - 1)$, the norm of this functional on $Z_{n,p}$ equals

$$\min_{P \in \mathcal{P}_{n-1}} \|H_g - P\|_q. \quad (6.4)$$

Here is a direct justification without a quotient-duality assumption. A minimizing polynomial exists by finite-dimensional coercivity and is unique for $q > 1$. Its residual $R = H_g - P$ satisfies

$$\int_0^1 |R|^{q-2} R Q = 0 \quad (Q \in \mathcal{P}_{n-1}). \quad (6.5)$$

If $R \neq 0$, the function $f = \text{sign}(R)|R|/\|R\|_q^{q-1}$ has L^p norm one and belongs to (6.2). It realizes the bound in Hölder's inequality. The zero-residual case is immediate. This proves (6.4).

For every fixed $p > 1$, the variation supremum is attained. A norm-bounded sequence is bounded in $W^{n,p}$ by successive integration, and its derivatives through order $n - 1$ are uniformly bounded and equicontinuous, using Hölder's inequality for the highest integral. A subsequence converges uniformly through order $n - 1$, and the highest derivatives have a weakly convergent subsequence in L^p . The limit satisfies all endpoint conditions, has norm at most one, and retains the limiting variation. The supremum is positive, so a maximizer must have norm exactly one by scaling.

For a norm-one variation maximizer u , choose $g = \text{sign}(u')$, with any value in $[-1, 1]$ on the zero set. The functional in (6.4) is at most $V_{n,p}$, since $\int g v' \leq \|v'\|_1$. On u it equals $V_{n,p}$. Thus its residual satisfies $\|R\|_q = V_{n,p}$, and equality in Hölder gives

$$u^{(n)}(t) = \text{sign } R(t) \left| \frac{R(t)}{V_{n,p}} \right|^{q-1}. \quad (6.6)$$

This formula gives more compactness than the original Sobolev bound.

Lemma 6 (compactness of maximizers). Suppose $p_j \rightarrow 2$ and u_j is any sequence of norm-one variation maximizers in X_{n,p_j} . Then, after a subsequence and a choice of global signs, u_j converges in $C^n[0, 1]$ to the positive norm-one Hilbert maximizer u_* .

Proof. Restrict to $3/2 \leq p_j \leq 5/2$, so $5/3 \leq q_j \leq 3$. The functions H_{g_j} , with $|g_j| \leq 1$, are uniformly bounded and Lipschitz. For $n = 2$, this follows directly from $H'_g = -g$; for $n \geq 3$, differentiate (6.3) once and bound its remaining integral.

The minimizing polynomials satisfy $\|P_j\|_{q_j} \leq 2\|H_{g_j}\|_{q_j}$. Their coefficients are uniformly bounded. To verify the needed uniform norm equivalence, a contrary sequence of polynomials can be normalized to have coefficient norm one and L^{q_j} norm tending to zero. A coefficient-convergent subsequence would give a nonzero polynomial vanishing almost everywhere, a contradiction. Consequently both P_j and their derivatives are uniformly bounded, and the residuals R_j are uniformly bounded and Lipschitz.

The quantities V_{n,p_j} are bounded above by (6.4) with the zero polynomial. They are bounded away from zero by testing the one fixed nonzero function u_* , whose highest derivative is bounded. Formula (6.6) now makes $u_j^{(n)}$ uniformly bounded and equicontinuous. For example, signed power maps with exponent in $[2/3, 2]$ on a fixed bounded interval have a common Hölder modulus with exponent $2/3$, after enlarging the constant. Arzelà--Ascoli supplies a uniformly convergent subsequence of the highest derivatives. Integration and the left endpoint conditions give convergence in C^n to some u .

Uniform boundedness and $p_j \rightarrow 2$ imply $\int |u^{(n)}|^2 = \lim_j \int |u_j^{(n)}|^{p_j} = 1$. All right endpoint conditions pass to the limit. Also $\|u'\|_1 = \lim_j V_{n,p_j}$. Testing $u_*/\|u_*^{(n)}\|_{p_j}$ gives $\liminf_j V_{n,p_j} \geq V_{n,2}$, while the Hilbert inequality applied to the limit gives the reverse bound. Thus u is a Hilbert maximizer. Theorem 1 identifies it with u_* or $-u_*$, proving the lemma. $\square$

6.2. Persistence of a single peak

We give the details that weak or uniform convergence alone would miss. The Hilbert midpoint representer, up to a positive factor, can be written

$$U(x) = F(|1 - 2x|), \quad F(z) = \int_z^1 (t^2 - z^2)^{n-1} dt, \quad 0 \leq z \leq 1. \quad (6.7)$$

To check this representation, expand the integrand. The resulting expression for F is an even polynomial of degree at most $2n - 2$, plus a multiple of z^{2n-1} . Thus (6.7) is polynomial of degree at most $2n - 1$ on each half-interval and has matching derivatives through order $2n - 2$ at the midpoint.

Near $z = 1$, writing $t = z + (1 - z)s$ shows that $F(z)$ is a positive constant times $(1 - z)^n$, to leading order. Therefore all endpoint derivatives below order n vanish. The jump of the derivative of order $2n - 1$ is nonzero. The weak derivative $(-1)^n D^{2n}U$ is a scalar multiple of the midpoint delta. Testing against U shows this scalar is positive because $\int |U^{(n)}|^2 > 0$ and $U(1/2) > 0$. Uniqueness of the Riesz representer proves the claim up to positive normalization.

For $n \geq 2$, differentiation gives

$$F'(z) = -2(n-1)z \int_z^1 (t^2 - z^2)^{n-2} dt < 0 \quad (0 < z < 1). \quad (6.8)$$

Also $F''(0) = -2(n-1)/(2n-3) < 0$. Hence $u'_* > 0$ on $(0, 1/2)$, $u'_* < 0$ on $(1/2, 1)$, and $u''_*(1/2) < 0$. The endpoint expansion gives $u_*^{(n)}(0) > 0$ and $(-1)^n u_*^{(n)}(1) > 0$.

These properties imply that every sufficiently close C^n perturbation satisfying the same clamped endpoint conditions has exactly one interior critical point, a strict maximum. Indeed, near zero its highest derivative remains positive and

$$u'(x) = \int_0^x \frac{(x-t)^{n-2}}{(n-2)!} u^{(n)}(t) dt > 0. \quad (6.9)$$

The reflected argument gives negativity near one. On compact intervals away from the endpoints and midpoint the strict derivative signs persist. On a small interval about the midpoint, the second derivative stays negative, so there is precisely one zero of the first derivative, bracketed by its signs on either side.

Lemma 6 therefore implies that for all p sufficiently close to two, every norm-one variation maximizer is, up to sign, positive and strictly unimodal. Such a maximizer has variation exactly twice its height. Since every admissible function has variation at least twice its height, this proves (6.1). It also proves coincidence of maximizing sets: a variation maximizer has height $C_{n,p}$, and every height maximizer has variation between $2C_{n,p}$ and $V_{n,p}$.

The only conclusion still needed for the local theorem is midpoint symmetry. Single-peak geometry by itself does not supply it.

6.3. Smoothness of the point-evaluation profile near two

Let $m = n - 1 \geq 2$, and define

$$K_a(t) = \frac{(a-t)_+^m}{m!}, \quad J(q, a) = \min_{P \in \mathcal{P}_m} \int_0^1 |K_a - P|^q dt. \quad (6.10)$$

By the same moment argument as (6.4), the evaluation norm is $J(q, a)^{1/q}$. We need uniform continuity on the whole interval in a , and twice continuous differentiability in a near $(q, a) = (2, 1/2)$.

First, minimizing polynomials have coefficients bounded uniformly for q in a compact subset of $(1, \infty)$ and $a \in [0, 1]$, by the argument used above. On this bounded coefficient set the integrands in (6.10) vary continuously with (q, a, P) , uniformly in t . Minimizing over that compact set proves continuity of J , including uniform convergence $J(q, \cdot) \rightarrow J(2, \cdot)$ as $q \rightarrow 2$. Strict convexity gives a unique minimizing polynomial; compactness and uniqueness prove continuous dependence of its coefficients.

The following elementary sublevel estimate handles exponents just below two. If P is a polynomial of degree at most m on an interval I of length bounded above and below, and $\|P\|_{L^\infty(I)} \geq b > 0$, then

$$|\{t \in I : |P(t)| \leq h\}| \leq C(m, b, I)h^{1/m} \quad (0 < h < 1). \quad (6.11)$$

The constant is uniform for intervals in the stated length range. To prove this, let the sublevel set have length $\ell > 0$. Divide its mass into sufficiently many equal portions and choose $m + 1$ points from alternate portions. They can be chosen in the sublevel set with consecutive separations bounded below by a fixed multiple of $\ell/(m + 1)$: coordinate separation is at least the measure of the intervening portion. Lagrange interpolation at these points gives $\|P\|_\infty \leq C_m h \ell^{-m}$, with the interval-length factor absorbed into the constant. Rearrangement gives (6.11); approximation of the chosen points handles essential rather than attained sublevel membership.

Consequently, for $0 < s < 1/m$, integration of distribution functions gives

$$\int_{\{|P| < h\}} |P|^{-s} dt \leq C h^{1/m-s}. \quad (6.12)$$

For example, split the region into dyadic bands $2^{-k-1}h \leq |P| < 2^{-k}h$, apply (6.11) to each, and sum the convergent geometric series. The estimates also show uniform integrability of these inverse powers.

At $(q, a) = (2, 1/2)$, the minimizing residual in (6.10) is a nonzero polynomial on each of the two half-intervals. If it were identically zero on one half, its orthogonality to constants would fail on the other: the remaining residual would be a nonzero constant multiple of the nonnegative truncated power. Continuity of the coefficients therefore supplies a uniform positive lower bound for the sup norm of each polynomial piece on its own interval when (q, a) and the polynomial coefficients are sufficiently close to this point. The intervals' lengths also stay bounded away from zero. Thus (6.11)–(6.12) apply uniformly to both pieces.

Write $P_c(t) = \sum_{j=0}^m c_j t^j$ and $R = K_a - P_c$. The normal equations are

$$\mathcal{N}_j(q, a, c) = \int_0^1 |R|^{q-2} R t^j dt = 0, \quad 0 \leq j \leq m. \quad (6.13)$$

Take a small open neighborhood with $q > 2 - 1/(2m)$. The equations are continuously differentiable in (q, a, c) there. Their derivatives in (a, c) are obtained by multiplying the variation of R by $(q - 1)|R|^{q-2}$, and the derivative in q has integrand $|R|^{q-2} R \log |R| t^j$, with value zero at $R = 0$. These assertions follow first away from a small sublevel set of R , by ordinary differentiation. The uniform bound (6.12) makes the omitted derivatives uniformly integrable and arbitrarily small as the sublevel threshold tends to zero. The logarithmic derivative is bounded by a constant times a small positive power of $|R|$ near zero. This also proves continuity of the derivatives by splitting into the same two regions. The moving interface $t = a$ causes no additional term, since K_a and its first a -derivative are continuous there for $m \geq 2$.

At the base point, the coefficient derivative of (6.13) is the negative definite Gram matrix $-\left[\int_0^1 t^{i+j} dt\right]_{i,j=0}^m$. The finite-dimensional implicit function theorem yields a continuously differentiable coefficient map $c(q, a)$. It is the minimizing polynomial map by strict convexity and continuity already established.

The envelope derivative and its second derivative are

$$\begin{aligned} J_a &= q \int |R|^{q-2} R K_a^{[1]}, \\ J_{aa} &= q(q-1) \int |R|^{q-2} (K_a^{[1]} - P_{c_a}) K_a^{[1]} + q \int |R|^{q-2} R K_a^{[2]}, \end{aligned} \quad (6.14)$$

where $K_a^{[1]} = (a - t)_+^{m-1}/(m-1)!$ and $K_a^{[2]} = (a - t)_+^{m-2}/(m-2)!$. For $m = 2$, the last expression is the indicator of $t < a$, with its value at $t = a$ immaterial. The coefficient term in the first derivative vanishes by (6.13). The second derivative exists and is continuous by (6.12), boundedness of these kernels, and dominated convergence away from the moving interface. In particular no claim that all residual roots are simple is needed.

At $q = 2$, the credited point-evaluation formula (4) gives

$$J(2, a) = \frac{[a(1-a)]^{2n-1}}{(2n-1)((n-1)!)^2}. \quad (6.15)$$

It has a unique maximum at $a = 1/2$, with strictly negative second derivative there. By continuity of (6.14), it follows that $J(q, \cdot)$ is strictly concave on one fixed neighborhood of $1/2$, for all q close enough to two. Uniform convergence on $[0, 1]$ forces every global maximum into that neighborhood. Reflection of the original evaluation problem gives $J(q, a) = J(q, 1-a)$. Strict concavity and this symmetry show that $1/2$ is the unique global maximizing point.

6.4. Completion of the proof

Shrink ε_n so that both the variation-maximizer argument and the evaluation-profile argument apply. We have established the constant identity and coincidence of maximizing sets. Every height maximizer must realize evaluation at its absolute maximum, and Section 6.3 forces that point to be $1/2$. The norm-one positive evaluation extremizer is unique by equality in Hölder in (6.4), or by strict convexity of the unit ball of L^p . Reflection preserves the norm and evaluation at $1/2$, so uniqueness makes this function midpoint symmetric. Section 6.2 already gives its strict derivative signs and its unique peak. This proves Theorem 5. $\square$

6.5. The unresolved global step

Compactness away from the endpoints of the exponent range. The argument in Lemma 6 gives the following more general statement. If $p_j \rightarrow p_0 \in (1, \infty)$ and u_j are norm-one variation maximizers, a subsequence converges in C^n to a norm-one variation maximizer at p_0 , and $V_{n,p_j} \rightarrow V_{n,p_0}$. To see this, keep p_j in any fixed compact subinterval of $(1, \infty)$; the exponents $q_j - 1$ then have a positive lower bound, so (6.6) still gives a common Hölder modulus. The limit has norm one at p_0 . For the lower bound on the limiting optimal value, test a maximizer at p_0 ; (6.6) makes its highest derivative bounded, so it is admissible at all nearby exponents and its varying norms converge. The upper bound follows by applying the p_0 inequality to the limit. This proves the statement without assuming uniqueness or symmetry at p_0 .

Thus a local continuation criterion at another exponent is available: if all its normalized variation maximizers are scalar-sign copies of one single-peak function with nonzero endpoint highest derivatives and a nondegenerate peak, and its point-evaluation profile has a unique midpoint maximum with a continuously persisting negative second derivative, the same argument proves the conjecture in a neighborhood of that exponent. These hypotheses must be established there; they cannot be imported from the case $p = 2$.

The strongest compactness assertion here is C^n convergence of *variation maximizers*, established from their dual optimality equation. It is not asserted for arbitrary normalized Sobolev sequences. This distinction is necessary to preserve the derivative signs near the clamped endpoints.

The point-profile calculation uses finite-dimensional best approximation only after proving uniform control of possible multiple zeros in its polynomial residual. Omitting (6.11)–(6.12) would leave the

two-sided neighborhood in p unjustified, since $q < 2$ creates negative powers at residual zeros.

This argument does not prove that the set of valid exponents is closed. At a hypothetical first exponent where the result ceases to hold, compactness of the old single-peak branch does not exclude a different multiple-peak global maximizer with the same value. Theorem 10 below resolves the separate point-evaluation symmetry question at every finite exponent. The possible coexistence of multiple-peak variation maximizers remains a substantive global obstacle.

The remaining task is to exclude such competing branches or to find a counterexample. Theorem 5 supplies a local result, not the full conjecture. No numerical experiment is used in its proof.

7. Complete symmetry at the fourth-order measure endpoint

We first settle the endpoint symmetry problem explicitly in order four by a finite alternation argument. The theorem below concerns the measure class (21); its constants also apply to the ordinary clamped Sobolev space by Theorem 4.

Theorem 8 (fourth-order endpoint). One has

$$C_{4,1} = \frac{1}{1296}, \quad V_{4,1} = \frac{1}{648}. \quad (7.1)$$

The norm-one relaxed extremizers for height and variation are exactly μ_* and $-\mu_*$, where

$$\mu_* = \frac{1}{9}(\delta_0 + \delta_1) - \frac{1}{4}(\delta_{1/6} + \delta_{5/6}) + \frac{5}{18}\delta_{1/2}. \quad (7.2)$$

Their potentials are even about $1/2$. No nonzero extremizer exists in the ordinary $W_0^{4,1}$ class. Thus the full constant, coincidence and symmetry assertions in [NS, Conjecture 4.14] hold for $n = 4$, $p = 1$, with the endpoint understood in the measure sense.

The all-order symmetry assertion is proved separately in Section 8. The point estimates for orders two and three are already included in [NS, Theorem 4.5]; those cases are not new here.

7.1. Extreme measures and a dual cubic

Let C be the maximal relaxed height. It is positive and attained by Section 5. Choose a maximizing point and sign. The corresponding evaluation functional exposes a nonempty compact face of $\mathcal{M}_4$. An extreme point of that face is extreme in $\mathcal{M}_4$ and still maximizes height. By the moment-circuit argument in Theorem 4, after choosing its positive potential it has nodes

$$\begin{aligned} 0 < a < b < c < 1, \quad (t_0, t_1, t_2, t_3, t_4) &= (0, a, b, c, 1), \\ \mu &= \frac{1}{D} \sum_{i=0}^4 \beta_i \delta_{t_i}, \quad \beta_i = \prod_{j \neq i} (t_i - t_j)^{-1}, \quad D = \sum_i |\beta_i|. \end{aligned} \quad (7.3)$$

The end nodes are 0 and 1 because stretching a shorter support, with derivative-measure norm fixed, multiplies height by the inverse cube of its length. The signs of the weights in (7.3) are $+, -, +, -, +$.

Write U for this positive potential and let x maximize it. Before a it is a positive multiple of x^3 ; after c it is a positive multiple of $(1 - x)^3$. Its derivatives at a and c have strict opposite signs. Hence $a < x < c$. Reflecting if necessary, assume

$$a < x \leq b < c < 1. \quad (7.4)$$

Put $K_x(t) = (x - t)_+^3/6$. There is a polynomial P of degree at most three such that

$$R(t) = K_x(t) - P(t), \quad \|R\|_\infty = C, \quad (R(0), R(a), R(b), R(c), R(1)) = (C, -C, C, -C, C). \quad (7.5)$$

Here are the duality details. The dual of $C[0, 1]/\mathcal{P}_3$ is the space of finite measures annihilating $\mathcal{P}_3$, with total-variation norm. Hahn-Banach and the Riesz representation theorem therefore identify the distance of K_x to $\mathcal{P}_3$ with the norm of its evaluation functional on $\mathcal{M}_4$. That norm equals C , because the chosen measure attains the global height. A best polynomial exists by finite-dimensional coercivity. Equality in $\int R d\mu \leq \|R\|_\infty \|\mu\|_{\text{TV}}$ forces the five values in (7.5). Since R is continuously differentiable, its derivative vanishes at a, b, c .

Let $R_+ = -P$, so that $R = R_+$ on $[x, 1]$. The cubic R_+ has a maximum at b and a minimum at c . Consequently, for some $k > 0$,

$$R'_+(t) = k(t - b)(t - c). \quad (7.6)$$

Equality of its values at b and 1 gives, on putting $h = c - b$,

$$1 - b = \frac{3}{2}h, \quad b = 1 - \frac{3}{2}h, \quad c = 1 - \frac{1}{2}h. \quad (7.7)$$

Indeed $\int_b^1 (t - b)(t - c)dt = (1 - b)^2((1 - b)/3 - h/2) = 0$.

7.2. The remaining algebraic constraints

On the left of x , $R(t) = R_+(t) + (x - t)^3/6$. The equations $R'(a) = 0$ and $R(a) = R(c)$ yield

$$(x - a)^2 = 2k(b - a)(c - a), \quad (x - a)^3 = k(c - a)^2(2(c - a) - 3h). \quad (7.8)$$

Set $v = (c - a)/h > 1$. Dividing the two equations, and using $0 < x - a \leq b - a$, gives

$$\frac{3}{2} < v \leq 2, \quad a = 1 - h\left(v + \frac{1}{2}\right), \quad x = 1 - \frac{h(2v - 1)}{2(v - 1)}. \quad (7.9)$$

For clarity, $(x - a)/h = v(2v - 3)/(2(v - 1))$. Its positivity gives $v > 3/2$, and its upper bound $v - 1$ gives $v \leq 2$.

The potential itself is stationary at x . Only the nodes $0, a$ contribute to its first derivative under (7.4), so $U'(x) = 0$ reads

$$(x - a)^2 = x^2 \frac{(b - a)(c - a)(1 - a)}{bc}. \quad (7.10)$$

Combining this with (7.8) gives $k = x^2(1 - a)/(2bc)$. Also $R(0) = R(1)$ implies

$$\frac{x^3}{6} = k \int_0^1 (t - b)(t - c)dt = \frac{kb^2}{3}, \quad x = \frac{b(1 - a)}{c}. \quad (7.11)$$

The integral equals $b^2/3$ by (7.7). All divided quantities are positive.

Substitute (7.7) and (7.9) into (7.10)-(7.11). The resulting two quadratic equations in h are

$$Ah^2 + Bh + C_0 = 0, \quad D_0h^2 + Eh + F = 0, \quad (7.12)$$

where

$$\begin{aligned} A &= 6v^2 - v - 4, & B &= -4v^2 - 4v + 6, \\ C_0 &= 4v - 4, & D_0 &= 2(v-1)(2v-1)^2, \\ E &= -16v^3 + 42v^2 - 31v + 4, & F &= 2v(2v-3)^2. \end{aligned} \quad (7.13)$$

One can check the second equation without elimination software: after substitution, the difference between the two sides of (7.10), multiplied by bc , is

$$(x-a)^2bc - x^2(b-a)(c-a)(1-a) = \frac{ah^2v}{8(v-1)^2}(D_0h^2 + Eh + F). \quad (7.14)$$

The first equation is $4(v-1)(xc - b(1-a)) = 0$. These identities justify (7.12) with no loss from a vanishing factor.

Two quadratics in (7.12) with a common root must satisfy

$$0 = (AF - C_0D_0)^2 - (AE - BD_0)(BF - C_0E) = 16(v-2)^2(v-1)(2v+1)\Pi(v), \quad (7.15)$$

where direct expansion gives

$$\Pi(v) = 8v^6 - 24v^5 - 6v^4 + 64v^3 - 48v^2 + 9v - 2. \quad (7.16)$$

To verify the first equality directly, put $L = AE - BD_0$, $M = AF - C_0D_0$ and $N = BF - C_0E$. Subtracting multiples of the two equations gives $Lh + M = 0$ and $h(Mh + N) = 0$. Since $h > 0$, these imply $M^2 - LN = 0$. The factorization in the second equality is an ordinary polynomial identity.

There is a short sign check for its last factor. For $z = v - 3/2 \in [0, 1/2]$,

$$\begin{aligned} \Pi(3/2 + z) &= 8z^6 + 48z^5 + 84z^4 + 28z^3 - \frac{87}{2}z^2 - 27z - 2 \\ &\leq -2z^2 - 27z - 2 < 0. \end{aligned} \quad (7.17)$$

Indeed the four positive terms are at most $(1/2 + 6 + 21 + 14)z^2 = (83/2)z^2$. Since $v > 3/2$, equation (7.15) forces $v = 2$. Both equations in (7.12) then reduce to $2(3h-2)(3h-1) = 0$. The root $h = 2/3$ makes $a < 0$ and is inadmissible. The other gives

$$(a, b, c, x) = \left(\frac{1}{6}, \frac{1}{2}, \frac{5}{6}, \frac{1}{2}\right). \quad (7.18)$$

Thus every globally maximizing extreme measure has precisely these nodes. This conclusion concerns maximizers directly; it does not require a knot-reflection inequality for every configuration.

7.3. Constants and all equality cases

The symmetric dual cubic, after normalization, is the centered cubic Chebyshev spline already described in [dB03, p. 74]. The new issue in this section is unrestricted global optimality and its equality classification. The weights in (7.3), normalized to total variation one at (7.18), are exactly (7.2). Their potential is

$$U_*(x) = \frac{1}{6} \left[\frac{x_+^3 + (x-1)_+^3}{9} - \frac{(x-1/6)_+^3 + (x-5/6)_+^3}{4} + \frac{5}{18}(x-1/2)_+^3 \right]. \quad (7.19)$$

It is symmetric and unimodal by Section 5. Substitution at $x = 1/2$ gives $U_*(1/2) = 1/1296$. The extreme maximizing measure exists, and (7.18) identifies it, so this value is the unrestricted sharp height, not merely the maximum within symmetric configurations. Theorem 4 then gives the variation constant $1/648$.

For an arbitrary norm-one height maximizer, expose a face by evaluation at one of its maximum-height points, with the appropriate sign. Every extreme point of that face is a height-maximizing extreme measure and therefore equals μ_* or $-\mu_*$. Only one of these two signs can belong to the chosen face, because the exposed value is positive. Krein-Milman makes the face a singleton. This proves the stated equality classification, and Theorem 4 transfers it to variation. An ordinary Sobolev extremizer would have an absolutely continuous fourth derivative, whereas (7.2) is atomic. Hence ordinary attainment is impossible. The approximation argument in Theorem 4 still recovers both sharp constants. $\square$

8. Symmetry in every order at the measure endpoint

The fourth-order calculation can be replaced by an all-order comparison. The key is to follow the contacts of the best uniform polynomial approximation as the evaluation point moves. Their velocities are strictly less than one. The monotonicity of B-spline peaks then compares the evaluation point with the peak of its own optimizing potential.

Theorem 9 (all-order measure endpoint). For every $n \geq 2$, the point-evaluation norm

$$A_n(x) = \sup_{\mu \in \mathcal{M}_n} U_\mu(x)$$

is strictly increasing on $(0, 1/2)$ and strictly decreasing on $(1/2, 1)$. There is a unique measure μ_n of total variation one attaining positive evaluation at $1/2$. Its potential is positive in $(0, 1)$, symmetric about $1/2$, and has exactly one peak. The normalized relaxed height and variation extremizers are precisely μ_n and $-\mu_n$. In particular, all assertions of [NS, Conjecture 4.14] hold at $p = 1$ for every order, with the endpoint interpreted in the measure sense. The sharp constants in the ordinary clamped Sobolev class agree with the relaxed constants, but no nonzero ordinary extremizer exists.

The full conjecture at the other exponents is not asserted here. The B-spline peak monotonicity used below is credited to Foucart [F]; the endpoint dependence needed here is supplied explicitly.

8.1. A zero count and regular contacts

We first record an elementary zero-count fact. Suppose f is polynomial of degree at most d on each side of a single knot x and belongs to C^{d-1} , with neither polynomial piece identically zero. Then the number of isolated zeros, counted with polynomial multiplicity away from x and the order of the first nonzero common derivative at x , is at most $d + 1$. The only case where no common derivative through order $d - 1$ is nonzero at a knot zero is

$$f(t) = a_-(t - x)^d \quad (t < x), \quad f(t) = a_+(t - x)^d \quad (t > x),$$

with $a_- a_+ \neq 0$; this case has just one distinct zero.

Here is a proof of the count, including missing coefficients. Write the common Taylor coefficients through degree $d - 1$ as $a_0, \dots, a_{d-1}$ and the two top coefficients as a_d^-, a_d^+ . If the first nonzero common coefficient has index r , factor $(t - x)^r$. Descartes' rule of signs bounds the zeros on the right and left by the sign variations in the remaining coefficient list and the list with signs alternating according to degree. For successive nonzero common coefficients whose degrees differ by h , the sum of the two variation contributions is at most h : it is one for odd h , and zero or two for even h . Their total is at most $d - r - 1$. The two independently chosen top coefficients add at most two. Adding the knot multiplicity r gives $d + 1$. If all common coefficients vanish, the displayed monomial case applies. Endpoint zeros of the interval under consideration are among the roots counted by this argument.

Fix $n \geq 3$, put $m = n - 1$, and write

$$K_x(t) = \frac{(x - t)_+^m}{m!}, \quad R_x(t) = K_x(t) - P_x(t), \quad A = A_n(x) = \|R_x\|_\infty, \quad (8.1)$$

where P_x is a best uniform polynomial of degree at most m . The moment duality in Section 7 applies in every order. A maximizing evaluation face has an extreme measure supported on $n + 1$ distinct nodes, and the signed weights force $n + 1$ alternating contacts of R_x .

Neither piece of R_x can be constant: smooth matching at x would make the other a constant plus a multiple of $(t - x)^m$, yielding at most two alternating contacts. Thus the contact set is finite. Every interior contact is a strict local extremum. Applying the zero count to R'_x with $d = n - 2$ shows that it has at most $n - 1$ zeros counting multiplicity. Since there are already at least $n - 1$ interior contacts, the contact set is exactly

$$0 = t_0(x) < t_1(x) < \dots < t_{n-1}(x) < t_n(x) = 1, \quad R_x(t_i) = (-1)^i A. \quad (8.2)$$

All interior critical points are simple, and the endpoint derivatives are nonzero. For $n = 3$, a contact cannot equal x : the two linear pieces of R'_x would otherwise have their common zero at x , and a second critical point would force one piece to vanish identically. For $n \geq 4$, a multiple critical point at x also violates the same zero count. Consequently, in every case,

$$\operatorname{sgn} R''_x(t_i) = (-1)^{i+1} \quad (1 \leq i < n), \quad R'_x(0) < 0, \quad \operatorname{sgn} R'_x(1) = (-1)^n. \quad (8.3)$$

Every measure maximizing positive evaluation at x is supported on this contact set, by equality in the dual norm inequality. Vandermonde rank gives a one-dimensional nullspace for the n moment conditions on $n + 1$ nodes. Positive evaluation and total variation one fix its sign and magnitude. Denote this unique measure by μ_x . Its potential is the positive normalized B-spline on these nodes.

The contacts, polynomial coefficients and A_n are C^1 functions of x . Indeed, differentiate the $n + 1$ contact-value equations and $n - 1$ critical-point equations. The contact-value block in the polynomial

coefficients and A is invertible: a nonzero polynomial of degree at most $n - 1$ cannot take nonzero alternating values at $n + 1$ ordered points. The remaining diagonal block consists of the nonzero numbers $R_x''(t_i)$. The implicit function theorem applies. Its equations are C^1 even when x equals a contact for $n \geq 4$; the case $n = 3$ was excluded above. To identify this local branch with the actual optimizer, use uniqueness and compactness: best polynomial coefficients are locally bounded and their subsequential limits are best approximants; the norming measures converge to the unique norming measure, whose $n + 1$ nonzero atoms prevent contact coalescence.

8.2. Every contact moves more slowly than evaluation

A dot denotes differentiation with respect to x ; a prime denotes differentiation with respect to t . Put $\ell = A_n'(x)/A_n(x)$ and define

$$V(t) = \dot{R}_x(t) + R_x'(t) - \ell R_x(t). \quad (8.4)$$

The identity $\partial_x K_x + \partial_t K_x = 0$ shows that V is again a degree-at-most- m spline with one simple knot: it is a polynomial minus ℓK_x . Differentiating (8.2) and using $R_x'(t_i) = 0$ at interior contacts gives

$$V(t_i) = 0 \quad (1 \leq i < n), \quad V(0) = R_x'(0) < 0, \quad V(1) = R_x'(1). \quad (8.5)$$

Neither piece of V vanishes identically, since its endpoint values are nonzero. The zero count gives at most n zeros counting multiplicity. The $n - 1$ prescribed interior zeros must therefore all be simple and there can be no additional zero. To see the parity step explicitly, the endpoint signs in (8.5) require the parity of the number of crossings to be $n - 1$. An additional simple zero or one double prescribed zero would reverse this parity; restoring it costs another zero and exceeds the bound n . A prescribed zero of multiplicity at least three already exceeds that bound. At a knot zero where the first m common coefficients vanish, both pieces would be monomials and there would be only one distinct zero, contrary to the $n - 1 \geq 2$ prescribed zeros. Thus this exception does not interfere with the parity argument.

It follows that $V'(t_i)$ has sign $(-1)^{i+1}$, the same as $R_x''(t_i)$. Differentiating the critical-point equation yields

$$\dot{R}_x'(t_i) + R_x''(t_i)t_i'(x) = 0, \quad V'(t_i) = R_x''(t_i)\{1 - t_i'(x)\}.$$

Hence

$$t_i'(x) < 1 \quad (1 \leq i < n). \quad (8.6)$$

No lower bound on these velocities is needed.

8.3. The dependence of a B-spline peak on all its knots

Let $B_{\mathbf{t}}$ be the conventional positive B-spline of degree $m \geq 2$ on $m + 2$ simple knots and let $s(\mathbf{t})$ be its unique peak. Foucart [F, Proposition 6] proves

$$\frac{\partial s}{\partial t_j} > 0 \quad \text{for every interior knot } t_j. \quad (8.7)$$

The peak is nondegenerate and depends differentiably on the knots. We need the same inequality for the two support endpoints. We prove this extension using the same divided-difference identities, keeping its scope separate from the cited statement.

By affine scaling take the support to be $[0, 1]$, and let B_j be the degree- $m + 1$ B-spline obtained by repeating the endpoint t_j . The knot-insertion identity in [F, Lemma 3 and Formulae 5] gives

$$B_j(y) = \frac{y - t_j}{m + 1} B'_j(y) + B_{\mathbf{t}}(y). \quad (8.8)$$

At $s = s(\mathbf{t})$, differentiating once and eliminating $B'_j(s)$ yields

$$(s - t_j)^2 B''_j(s) = m(m + 1) \{B_j(s) - B_{\mathbf{t}}(s)\}. \quad (8.9)$$

For the left endpoint $t_j = 0$, integrating (8.8) with $B_j(1) = 0$ gives

$$B_j(s) = (m + 1)s^{m+1} \int_s^1 B_{\mathbf{t}}(y) y^{-m-2} dy \leq B_{\mathbf{t}}(s)(1 - s^{m+1}) < B_{\mathbf{t}}(s). \quad (8.10)$$

For the right endpoint, the reflected formula gives the same strict inequality. Thus $B''_j(s) < 0$. Differentiation of the divided difference defining $B_{\mathbf{t}}$ with respect to an endpoint gives

$$\partial_{t_j} B'_{\mathbf{t}}(s) = -\frac{1}{m + 1} B''_j(s) > 0. \quad (8.11)$$

The derivative of the support-length normalization contributes a multiple of $B'_{\mathbf{t}}(s) = 0$, so it contributes nothing to (8.11). Since $B''_{\mathbf{t}}(s) < 0$, implicit differentiation of $B'_{\mathbf{t}}(s) = 0$ proves (8.7) for the endpoints too.

Translation equivariance now gives

$$\sum_{j=0}^{m+1} \frac{\partial s}{\partial t_j} = 1, \quad \frac{\partial s}{\partial t_j} > 0 \quad \text{for every } j. \quad (8.12)$$

8.4. Strict midpoint comparison and equality

Let $s(x) = s(t_0(x), \dots, t_n(x))$ be the peak of U_{μ_x} . The normalizing factor of this potential does not affect its peak. Equations (8.6) and (8.12), together with $t'_0 = t'_n = 0$, show that

$$s'(x) = \sum_{i=1}^{n-1} \frac{\partial s}{\partial t_i} t'_i(x) < \sum_{i=1}^{n-1} \frac{\partial s}{\partial t_i} < 1. \quad (8.13)$$

Reflection and uniqueness of the positive-evaluation norming measure at $1/2$ give $s(1/2) = 1/2$. Therefore $s(x) - x$ is strictly decreasing and

$$s(x) > x \quad (x < 1/2), \quad s(x) < x \quad (x > 1/2). \quad (8.14)$$

Finally, uniqueness and weak-star continuity of μ_x , together with joint continuity of $\partial_x K_x$ for $n \geq 3$, give the envelope identity

$$A'_n(x) = U'_{\mu_x}(x). \quad (8.15)$$

It follows directly by testing the optimizer at each of two neighboring points in the other evaluation problem and taking the two difference quotients. The B-spline potential is strictly increasing before its unique peak and strictly decreasing after it. Equation (8.14) therefore proves the asserted strict monotonicity of A_n .

Every global height maximizer must consequently attain its absolute height at $1/2$. The unique norming measure there, up to the sign of evaluation, is μ_n . Its potential is symmetric by reflection and positive by the moment-circuit formula. Theorem 4 transfers the complete classification to variation. The ordinary constants agree by the recovery argument there; an ordinary extremizer would have an absolutely continuous highest derivative, whereas the unique relaxed one is a nonzero atomic circuit. This rules out ordinary attainment.

For $n = 2$, the direct formula $A_2(x) = x(1 - x)/2$ and its unique three-point norming measure give the same conclusions. This completes the proof. $\square$

9. Point symmetry at every finite exponent

The preceding cancellation also has a finite-exponent form. Instead of following finitely many contacts, we integrate an auxiliary spline against a positive weight and count its zeros. Throughout this section $q = p/(p - 1)$ is the dual exponent; it is not the target exponent in [NS, Conjecture 4.3].

Theorem 10 (finite-exponent point symmetry). Let $n \geq 2$ and $1 < p < \infty$. The point-evaluation norm

$$A_{n,p}(a) = \sup\{u(a) : u \in W_0^{n,p}(0, 1), \|u^{(n)}\|_p \leq 1\} \quad (9.1)$$

is strictly increasing on $(0, 1/2)$ and strictly decreasing on $(1/2, 1)$. The nonzero extremizers for the height inequality in (2) are precisely the scalar multiples of the unique positive norm-one midpoint normer. That normer is symmetric, positive in the interior, and has a unique nondegenerate maximum.

The order-two statement and order-three midpoint symmetry are prior results [W, Theorem 1.1 and Lemma 1.3], recorded also in [NS, Theorem 4.5]. Watanabe et al. explicitly note in [W, Remark 1.4] that their symmetrization proof does not extend to orders at least four. The theorem above gives the all-order comparison. Together with Theorem 9 and the credited $p = \infty$ theorem [NS, Theorem 4.12], it establishes the $k = 0$, target- L^∞ case of [NS, Conjecture 4.3] for every $n \geq 2$ and $1 \leq p \leq \infty$. It does not establish the remaining variation identity in Conjecture 4.14.

9.1. Simple residual zeros and a single peak

We prove the theorem for $n \geq 3$ and write $m = n - 1$. As in Section 6, let P_a be the unique best L^q polynomial approximant of degree at most m to $K_a(t) = (a - t)_+^m/m!$. Set

$$R_a = K_a - P_a, \quad f_a = |R_a|^{q-2}R_a, \quad u_a(t) = \int_0^t \frac{(t - z)^{n-1}}{(n - 1)!} f_a(z) dz. \quad (9.2)$$

The normal equations (6.5) make u_a clamped at both endpoints and give

$$u_a(a) = \int_0^1 |R_a|^q > 0, \quad v_a = \frac{u_a}{\|R_a\|_q^{q-1}}. \quad (9.3)$$

Here v_a is the unique positive-evaluation normer. This follows from the dual formula and equality in Hölder's inequality, as in Section 6; the best-approximation formula itself is credited to [GS, Theorem 2 and Corollary 1].

The residual has at least n sign changes. Otherwise the polynomial having its sign-change points as roots, with a suitable sign, has degree at most $n - 1$ and gives a strictly positive integral against f_a , contradicting (6.5). Neither polynomial piece of R_a vanishes identically: smooth matching would make the other a nonzero multiple of $(t - a)^m$, of constant sign on its side, contradicting the constant moment. The one-knot zero count of Section 8.1 bounds the total number of zeros, with multiplicity, by n . Its exceptional two-monomial case has only one distinct zero and is excluded. Thus R_a has exactly n simple interior zeros, and its endpoint values are nonzero.

Repeated Rolle counting shows that u_a has no interior zero and u'_a has exactly one simple interior zero. Indeed, $u_a \in C^n$ has endpoint zeros of multiplicity n . An additional interior zero would force at least $n + 1$ distinct zeros of f_a after n differentiations. Applied to u'_a , whose endpoint multiplicities are $n - 1$, the same count excludes either two distinct interior zeros or a degenerate zero. Consequently $u_a > 0$ in $(0, 1)$ and its unique peak $s(a)$ satisfies

$$u''_a(s(a)) < 0, \quad f_a(0) > 0, \quad (-1)^n f_a(1) > 0. \quad (9.4)$$

The last two signs follow from the endpoint Taylor expansions.

9.2. Differentiation through the residual zeros

The preceding simple-root result removes the restriction to exponents near two in the regularity argument. Fix $q > 1$ and $a \in (0, 1)$. The residual is C^1 in t , since $m \geq 2$. Its roots are transverse even if one equals the kernel knot. In a small neighborhood of the minimizing polynomial coefficients and evaluation point, there remain exactly n simple roots with inverse slopes uniformly bounded. For $1 < q < 2$, this gives the locally uniform estimate

$$\int_{\{|R| < h\}} |R|^{q-2} dt \leq Ch^{q-1}. \quad (9.5)$$

It follows by changing variables from t to $R(t)$ in disjoint monotone neighborhoods of the roots; outside them $|R|$ is bounded below. For $q \geq 2$ the weight has no singularity. Differentiating the normal equations in a and the polynomial coefficients is therefore justified by the same sublevel splitting as in Section 6.3. The coefficient Jacobian is

$$-(q - 1) \left[\int_0^1 |R_a|^{q-2} t^{i+j} dt \right]_{i,j=0}^{n-1}, \quad (9.6)$$

a negative definite Gram matrix. The implicit function theorem and uniqueness give a C^1 minimizing coefficient map for every fixed $q > 1$.

The same estimate shows that $a \mapsto f_a$ is continuously differentiable as a map into L^1 . Its derivative is $(q - 1)|R_a|^{q-2} \dot{R}_a$. As a function of t , f_a is absolutely continuous, with derivative $(q - 1)|R_a|^{q-2} R'_a$. Integration gives continuous differentiability of $a \mapsto u_a$ into C^{n-1} , as well as $u_a \in W^{n+1,1}$. The nondegenerate peak in (9.4) therefore depends continuously differentiablely on a . In what follows, a dot denotes the a derivative and a prime the t derivative.

9.3. A weighted comparison for peak motion

Fix a and abbreviate $u = u_a$, $R = R_a$, $f = f_a$, and $s = s(a)$. Define

$$\ell = \frac{\dot{u}(s)}{u(s)}, \quad W = \dot{u} + u' - \ell u, \quad H = \dot{R} + R'. \quad (9.7)$$

The identity $\partial_a K_a + \partial_t K_a = 0$ makes H a polynomial of degree at most $n - 1$. Almost everywhere,

$$D^n W = |R|^{q-2} T, \quad T = (q - 1)H - \ell R. \quad (9.8)$$

The function T is a degree-at-most- $n - 1$ spline with one simple knot. If both pieces are nonzero, Section 8.1 bounds its isolated zeros by n , with the two-monomial exception having only one distinct zero. The weight in (9.8) is positive almost everywhere and integrable.

The function $W \in W^{n,1} \subset C^{n-1}$ satisfies

$$\begin{aligned} W^{(j)}(0) &= W^{(j)}(1) = 0 & (0 \leq j \leq n - 2), \\ W^{(n-1)}(0) &= f(0) > 0, \quad W^{(n-1)}(1) = f(1), \\ W(s) &= 0. \end{aligned} \quad (9.9)$$

In particular, W is positive near zero and negative near one. We claim

$$W'(s) < 0. \quad (9.10)$$

Here is a zero-count proof that includes the possible singularities of the weight. If $W'(s) > 0$, the opposite endpoint signs force another root on each side of s . If $W'(s) = 0$ and $W''(s) \neq 0$, there is a double root and another root. If $W'(s) = W''(s) = 0$, there is a root of multiplicity at least three. These derivatives exist because $n \geq 3$. Thus failure of (9.10) supplies interior zeros of total multiplicity at least three, besides the two endpoint zeros of multiplicity $n - 1$.

Apply ordinary Rolle's theorem $n - 1$ times. It yields at least $n + 2$ distinct zeros of $W^{(n-1)}$: the initial count is at least $2n + 1$, and all retained multiplicities are exhausted by this stage. On each of the $n + 1$ intervening disjoint intervals, the integral of $|R|^{q-2} T$ is zero. If both pieces of T are nonzero, it is not identically zero on any interval. Positivity of the weight then forces a sign change of T on each interval, contradicting its zero bound.

If one piece of T vanishes identically, smooth matching makes the other a constant multiple of $(t - a)^{n-1}$. Then $W^{(n-1)}$ is the nonzero constant $f(0)$ or $f(1)$ on the zero side and is strictly monotone on the other side. It has at most one zero, again a contradiction. If both pieces vanish, it is constant nonzero and has none. This proves (9.10).

Differentiate $u'_a(s(a)) = 0$. Equations (9.7) and (9.10) give

$$W'(s) = u''_a(s) \{1 - s'(a)\} < 0. \quad (9.11)$$

Since $u''_a(s) < 0$, we obtain $s'(a) < 1$ for every $a \in (0, 1)$.

9.4. Strict midpoint maximization and the remaining gap

Reflection and uniqueness give $s(1/2) = 1/2$. Thus $s(a) - a$ is strictly decreasing, positive before the midpoint and negative after it. The envelope identity is

$$A'_{n,p}(a) = \frac{u'_a(a)}{\|R_a\|_q^{q-1}}. \quad (9.12)$$

The strict single-peak signs of u_a prove the stated monotonicity. Any height extremizer must attain its absolute maximum at $1/2$; uniqueness of the midpoint normer gives all equality cases, and reflection gives its symmetry. This proves Theorem 10. $\square$

Two useful boundary identities follow from the same cancellation. With $J(a) = \int_0^1 |R_a|^q$, integration by parts and the normal equations give

$$\begin{aligned} J'(a) &= |R_a(0)|^q - |R_a(1)|^q, \\ ((n-1)q+1)J(a) &= a|R_a(0)|^q + (1-a)|R_a(1)|^q. \end{aligned} \quad (9.13)$$

For the second identity use $(a-t)R'_a = -mR_a$ plus a polynomial of degree at most m . The theorem therefore also proves the strict ordering of the endpoint residual magnitudes on either side of the midpoint.

This argument classifies height extremizers for every finite exponent. It does not imply that every variation maximizer has a single peak. Its dual residual in (6.6) comes from H_g , which may have many changes of polynomial piece when $g = \text{sign}(u')$. The one-knot bound in (9.8) is then unavailable. This is the remaining obstruction to the full variation conjecture.

10. Variation continuation from the measure endpoint

The unique endpoint measure also controls arbitrary variation maximizers as the input exponent decreases to one. Two issues require care: ordinary highest derivatives develop concentration, and in order three weak measure convergence does not directly give uniform second-derivative convergence.

Theorem 11 (continuation above one). For every $n \geq 3$ there exists $\eta_n > 0$ such that, for $1 < p < 1 + \eta_n$,

$$V_{n,p} = 2C_{n,p}. \quad (10.1)$$

The nonzero extremizers in the two inequalities are exactly the scalar multiples of the unique positive norm-one midpoint normer. They are symmetric and strictly unimodal up to sign. The interval width is not estimated, and overlap with the interval in Theorem 5 is not asserted. The case $n = 2$ is already known for all p by [NS, Remark 4.15].

10.1. Compactness of arbitrary endpoint sequences

Fix $n \geq 3$. Let $p_j > 1$ tend to one and let u_j be any norm-one variation maximizers, whose existence was proved in Section 6. Write $f_j = u_j^{(n)}$ and $\mu_j = f_j dt$. Since the interval has length one,

$$\|\mu_j\|_{\text{TV}} = \|f_j\|_1 \leq \|f_j\|_{p_j} = 1.$$

A subsequence converges weak-star to some $\mu \in \mathcal{M}_n$. For $n \geq 3$ the potential kernel and its first derivative are jointly continuous on the square. Their integrals over the unit measure ball are uniformly equicontinuous. Pointwise weak-star convergence and this equicontinuity therefore give $u_j \rightarrow U_\mu$ in $C^1[0, 1]$.

The constants satisfy $V_{n,p_j} \leq V_{n,1}$. Conversely, given $\delta > 0$, the recovery argument in Theorem 4 supplies one nonzero smooth clamped function v with

$$\frac{\|v'\|_1}{\|v^{(n)}\|_1} > V_{n,1} - \delta.$$

Its highest-derivative L^{p_j} norm tends to its L^1 norm. Testing this fixed function and then letting δ decrease to zero proves $V_{n,p_j} \rightarrow V_{n,1}$. Thus U_μ maximizes endpoint variation. Theorems 4 and 9 identify μ with μ_n or $-\mu_n$. After choosing the common sign along the subsequence, we have

$$\mu_j \xrightarrow{*} \mu_n, \quad u_j \rightarrow U_{\mu_n} \quad \text{in } C^1, \quad \|\mu_j\|_{\text{TV}} \rightarrow 1. \quad (10.2)$$

The last limit follows from lower semicontinuity and the upper bound one.

We also need convergence of the absolute measures:

$$|\mu_j| \xrightarrow{*} |\mu_n|. \quad (10.3)$$

Indeed, any weak-star limit ν of a subsequence of $|\mu_j|$ dominates $|\mu_n|$. To see this, for nonnegative continuous h and continuous ϕ with $|\phi| \leq h$, pass to the limit in $|\int \phi d\mu_j| \leq \int h d|\mu_j|$ and use the dual description of variation. Both ν and $|\mu_n|$ have mass one by (10.2), so they agree. Every subsequence has this same limit, proving (10.3).

10.2. The limiting dual residual fixes the endpoint signs

Put $U = U_{\mu_n}$ and let $g_j = \text{sign}(u'_j)$, with any choice in $[-1, 1]$ where $u'_j = 0$. The strict signs of U' imply pointwise convergence except at $0, 1/2, 1$ to the midpoint sign function g_* , which is positive before $1/2$ and negative after it. Dominated convergence gives $g_j \rightarrow g_*$ in L^1 . Hence $H_j = H_{g_j}$ converges uniformly to

$$H_*(t) = 2K_{1/2}(t) - \frac{(1-t)^{n-1}}{(n-1)!}. \quad (10.4)$$

Let P_j minimize $\|H_j - P\|_{q_j}$ over polynomials of degree at most $n-1$, where $q_j = p_j/(p_j-1) \rightarrow \infty$. The bound $\|P_j\|_1 \leq \|P_j\|_{q_j} \leq 2\|H_j\|_\infty$ and finite-dimensional norm equivalence give bounded polynomial coefficients. Take any subsequential coefficient limit P_∞ . For each fixed finite r , each polynomial P in the same space, and all sufficiently large j ,

$$\|H_j - P_j\|_r \leq \|H_j - P_j\|_{q_j} \leq \|H_j - P\|_{q_j} \leq \|H_j - P\|_\infty. \quad (10.5)$$

First let j tend to infinity, using uniform convergence, and then let r tend to infinity. Thus P_∞ is a best uniform approximant to H_* . Uniqueness of best uniform polynomial approximation and (10.4) identify

$$R_j := H_j - P_j \rightarrow 2R_{1/2} \text{ quadratically uniformly,} \quad f_j = \text{sign}(R_j) \left| \frac{R_j}{V_{n,p_j}} \right|^{q_j-1}. \quad (10.6)$$

Here $R_{1/2}$ is the truncated-power residual in Section 8; every subsequence has the same limit. The second identity is the dual optimality equation (6.6).

The endpoint residual values in (8.2) are nonzero, positive at zero and of sign $(-1)^n$ at one. Uniform convergence therefore fixes these signs for R_j on small endpoint neighborhoods, independently of j once it is large. Equation (10.6) fixes the same strict signs for f_j , although its magnitude need not remain bounded. Integration from the clamped endpoints gives $u'_j > 0$ near zero and $u'_j < 0$ near one. On compact sets separated from the endpoints and midpoint, uniform convergence of u'_j to U' preserves the required signs as well.

10.3. Curvature at the midpoint and completion

For $n \geq 4$, the twice-differentiated potential kernel is jointly continuous. The compactness argument above then gives $u''_j \rightarrow U''$ uniformly. The B-spline peak is nondegenerate, as used in Section 8 and [F, Proposition 6]. Thus $u''_j < 0$ on a fixed central interval for all sufficiently large j .

For $n = 3$, the unique symmetric measure has four distinct knots $0, a, 1 - a, 1$, with $a < 1/2$. The midpoint lies in a knot-free interval, where U'' is a strictly negative constant. Choose a closed central subinterval $I = [c, d]$ entirely inside this gap. Equation (10.3) and $|\mu_n|(I) = 0$ imply $|\mu_j|(I) \rightarrow 0$. Also $\mu_j([0, c]) \rightarrow \mu_n([0, c])$: continuous approximation of the interval indicator, together with (10.3), justifies this passage because its relative boundary $\{c\}$ has zero limiting variation mass. The endpoint atom at zero is included on both sides. Since $u''_j(x) = \mu_j([0, x])$, we obtain

$$\begin{aligned} \sup_{x \in I} |u''_j(x) - U''(x)| &\leq |\mu_j([0, c]) - \mu_n([0, c])| + |\mu_j|(I) \\ &\longrightarrow 0. \end{aligned} \tag{10.7}$$

Hence the middle curvature is strictly negative in order three too.

Combine this curvature with the derivative signs already proved near the endpoints and on the intervening compact intervals. Each u_j has exactly one critical point for large j , with positive derivative before it and negative derivative after it. Because the original sequence consisted of arbitrary variation maximizers, a contradiction-sequence argument supplies $\eta_n > 0$ for which every normalized maximizer has this geometry, up to sign, whenever $1 < p < 1 + \eta_n$.

Such a maximizer has variation twice its height, so $V_{n,p} \leq 2C_{n,p}$. The reverse bound always holds. A variation maximizer is therefore a height maximizer, and a height maximizer is a variation maximizer. Theorem 10 gives the midpoint symmetry and unique equality class. This proves Theorem 11. $\square$

11. Remaining questions

The full all-exponent form of [NS, Conjecture 4.14] remains unresolved. Theorem 9 completes its measure endpoint. Theorems 5 and 11 give order-dependent exponent intervals around two and immediately above one, respectively. Their widths are not estimated, and their overlap is not established. Theorem 10 settles midpoint symmetry and height-extremizer uniqueness for every finite exponent, completing the $k=0$, target- L -infinity slice of Conjecture 4.3 together with endpoint results. What remains for Conjecture 4.14 is the variation identity and exclusion of competing multiple-peak variation maximizers at the other exponents. Compactness of a known branch alone does not exclude coexistence with such a maximizer. The stronger centroid and knot-reflection inequalities in the research notes remain unproved and are not needed for Theorems 9–11.

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